

CARSEM

ENGINEERING CHANGE NOTICE

E.C.N. NO : 2512018M

PAGE 1 OF 163

TITLE : CONTROL OF NON-CONFORMING MATERIAL/PRODUCTS

SPECIFICATION NO : CRQA100H07

REVISION : CN

☒ PERMANENT	☐ NEW SPECIFICATION
☒ SPECIFICATION REVISION	
☐ REVIEW FOR COMPLIANCE	
☐ OBSOLETE SPECIFICATION	

☐ TEMPORARY ECN	☐ *PERMANENT CHANGE
☐ TEMPORARY CHANGE	

*PERMANENT CHANGE TO BE ENCLOSED WITH ECN TO SPEC DEPARTMENT THROUGH E-MAIL.

TRAINING NEEDS:

(TO BE FILLED BY ORIGINATOR)

☒	ORIGINATOR REQUIRE UPDATE TRAINING MANUAL INFORMATION AND CONDUCT TRAINING
☐	TRAINING NOT REQUIRED
☐	TRAINING TO BE DONE BY TRAINING DEPT AND REQUIRE TO UPDATE THE TRAINING MANUAL
☐	JUSTIFICATION IF NO REQUIRE TO UPDATE TRAINING MANUAL CHANGES NOT RELATED TO THE PROCESS

TRAINING DEPT'S ACKNOWLEDGEMENT : **MAHANI / KARIMUDDIN / CHEN XH**

JUSTIFICATION FOR CHANGE :

NOTE : - THOSE MARKED * MUST REQUIRE ECN TO BE RAISED OTHERWISE TECN WILL BE REJECTED.
EXCEPTION IS CONSIDERED PROVIDED EVIDENCE ATTACHED FOR THOSE ITEM MARKS '*' ONLY.

☐ * QUALIFICATION BUILD	☐ AUDIT RESPONSE	☐ NEW MACHINE / PACKAGE / MATERIAL / PROCESS
☐ NEW SPECIFICATION / PROCEDURE	☐ PROCESS / QUALITY IMPROVEMENT	☒ 8D ACTIONS, CARSEM REF # : SRENC2025113526
☐ *EVALUATION / RISK BUILD	☐ SPECIFICATION COMPLIANCE	☐ 3D ACTIONS, CARSEM REF # : _____
☐ OTHERS, PLEASE SPECIFY : _____		

CHANGE TYPE :

☐ CUSTOMER REQUEST
☒ CARSEM INITIATED

THE SPECIFICATION IS APPLICABLE TO :

▪ CARSEM (M) :	☒ M-SITE	☒ S – SITE
▪ CARSEM CHINA :	☒ SUZHOU SITE	☒ SUXIANG SITE

TIME FRAME (FOR TECN ONLY)- PLEASE SPECIFY :

_____ WEEK / S or _____ MONTHS / S
TECN FOR MORE THAN 1 MONTH CAN BE ALLOWED FOR EVALUATION PURPOSE ONLY

EXPIRY DATE :					
☐ NEW	☐ EXTENSION, _____ TIME				

THE CHANGE(S) IS APPLICABLE TO (FOR TECN ONLY):

▪ CARSEM (M) :	☐ M – SITE	☐ S – SITE
▪ CARSEM CHINA :	☐ SUZHOU SITE	☐ SUXIANG SITE
** COUNTER-PART APPROVAL	☐ REQUIRED	☐ NOT REQUIRED
** CHANGES REQUIRED PCN	☐ YES	☐ NO
IF YES , PCN NUMBER : _____		

NOTE: ** DETERMINED BY ORIGINATOR

DESCRIPTION OF CHANGE :

FROM : N/A

TO : AS PER HISTORY OF CHANGES (PAGE 163 OF 163)

RELATED/CROSS REFERENCE SPEC TO BE REVISED:

☒ NO
☐ YES. SPEC NO. _____

ORIGINATOR	LAILY AZWATI							
INITIATOR	SK LEE							
QA ENGINEER	SHANKRI	SUFIAN	KW LEE	CHANTHIRAVANAN	SURESH	KALAI	AZIZAH	XIA CAIHUA
PROCESS ENGINEER	JUDE	MK KAN	ROSLI	SARAVANAN	YY CHIN	CL CHAN	KF LOH	
	KW YEE	CL WONG	XIA RONGGUI	WANG MINGREN				
ENGINEERING MANAGER	ZAMRI	NAVINDRAN	IRWAN	RICHARD LEE	KF WONG	CHRIS WONG	SHUKRI NIZAM	
	YK YANG	ANDY XIANG						
QA MANAGER	RITA ANN	SY LUM	KT HUM	THAVAKUMARAN	YORK XIAO			
QMS MANAGER	REETHA							
HEAD OF QUALITY	WAYNE LAI							

(TO BE FILLED BY ORIGINATOR)

CUSTOMER APPROVAL :	☐ REQUIRED	☒ NOT REQUIRED
COMPANY NAME :	NAME / SIGNATURE	

APPROVAL FOR SPECIFICATION RELEASE : SK LEE/ LAILY AZWATI

EFFECTIVE DATE : **2026-JAN-29**

PROPRIETARY INFORMATION DO NOT REPRODUCE OR FURTHER DISCLOSE WITHOUT AUTHORIZATION FROM CARSEM.
ALL PRINTED COPY WILL BE CLASSIFIED AS UNCONTROLLED DOCUMENT UNLESS WITH CARSEM DOCUMENT CONTROL STAMP IN RED INK

FORM # CRQA100G01 – 1 REV : R RETENTION REQUIRED

CARSEM	PROPRIETARY INFORMATION. DO NOT REPRODUCE OR FURTHER DISCLOSE WITHOUT AUTHORIZATION FROM CARSEM	SPEC NO.	CRQA100H07
		REV.:	CN
		PAGE :	2 OF 163

- 1.0 Title
Control of Non-conforming Material/Products
- 2.0 Purpose:
To establish and provide guideline and general procedure for quarantine/on-hold, recapture and MRB whenever non-conforming material/products are found.
- 3.0 Scope:
Apply to all assembly and test areas.
- 4.0 Reference:
 - 4.1 CRQA100G02 - Document/Records Retention Procedure.
 - 4.2 CRQA100M01 - Corrective and Preventive Action Procedure.
 - 4.3 CRQA100N02 - Customer Complaint and Product Return Procedure.
 - 4.4 QRA000002 – Carsem’s Rework Policy.
 - 4.5 QRA000024 – Lot Tolerance Defective (LTPD) Sampling Plan.
 - 4.6 QCG000080 –Guideline for the handling of Internal & Customer Concession Permit.
 - 4.7 CRQA100H03 – Incoming Material Quality Control.
 - 4.8 QCG-073 – Product Scrap page Procedure.
 - 4.9 QCG000017 – Guideline for a Shutdown of an Operation due to Quality Issue.
 - 4.10 IQC00WK02 – Standard Operating Procedure for field failure feedback report to IQA.
 - 4.11 CRQA100H02- Handling, Storage, Packing, Preservation and Delivery.
 - 4.12 QCG000112 - Quality abnormality Management System for Critical Defect.
 - 4.13 IATF 16949:2016 – Automotive Quality Management System Standard.
 - 4.14 ISO 9001:2015 – Quality Management Systems Requirements.
 - 4.15 EMS000024 – Hazardous Substances Management System Manual.
- 5.0 Responsibility:
 - 5.1 Incoming Personnel:
 - 5.1.1 IQA Inspector:
 - To assist IQA supervisor to do physical recapture and containment.
 - To perform screening as required.
 - 5.1.2 IQA Supervisor:
 - To quarantine and hold non-conforming material in system.
 - To ensure recapture and containment of relevant lots are done and result feedback to relevant personnel (e.g. Supplier /IQA Engineer/ Process Engineer).
 - 5.1.3 IQA Engineer:
 - To verify defect and also determine sampling plan for recaptured material.
 - To coordinate recapture activity of non-conforming product/ material.
 - To conduct risk assessment
 - To raise a MRB for Material disposition (when the problem was detected at IQA) if customer’s waive / disposition required.
 - To notify and issue Supplier Corrective Action Request (SCAR) to related supplier.

CARSEM	PROPRIETARY INFORMATION. DO NOT REPRODUCE OR FURTHER DISCLOSE WITHOUT AUTHORIZATION FROM CARSEM	SPEC NO.	CRQA100H07
		REV.:	CN
		PAGE :	3 OF 163

5.2 Line Personnel:

5.2.1 Operator / Technician

- To notify production supervisor or process technician immediately upon detection of any non-conforming products.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16 ,17,18 and 19).

5.2.2 Production Supervisor

- To confirm/notify QA immediately upon confirmation of the non-conforming material.
- To submit the non-conforming material/products to QA to quarantine the parts at QA Quarantine Cabinet.
- To notify production control and process engineering that the non-conforming material was detected and placed on hold by QA.
- To verify and confirm the non-conformance status of the said product/material.
- To assist QA in carrying out recaptures based on Engineering Recapture instruction and Recapture procedure in the specification.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16 ,17,18 and 19).

5.2.3 Production Superintendent

- Coordinate and ensure Recapture of product has been carried out effectively.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19).

5.2.4 Maintenance Superintendent

- To perform machine and product buy-off with process/ product engineer before released machine for production.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15,\ 16 ,17,18 and 19).

5.2.5 Manufacturing/Product Manager

- To approve the disposition made in MRB after reviewing all pertinent support documents.

5.3

QA Personnel

5.3.1 QA Inspector

- Issue Non-Conformance Report (NCR) to production supervisor if any defects detected at QA Gate.
- To stop QA Gate and notify QA Supervisor/Engineer if found critical defects and 3 times rejection (2 times for automotive product in SZ Site) of the same lot.

5.3.2 QA supervisor / Leader

- To quarantine all affected/ recaptured non conforming material at the QA quarantine area and hold electronically in MES system.
- To coordinate with production to carry out the recapture based on Engineering Recapture instruction and Recapture procedure in the specification.
- To ensure all affected sub-lots are physically pulled back by performing verification of actual on-hold lot # and lot quantity (tally check on system hold report versus original on-hold request list) to ensure no miss out.

CARSEM	PROPRIETARY INFORMATION. DO NOT REPRODUCE OR FURTHER DISCLOSE WITHOUT AUTHORIZATION FROM CARSEM	SPEC NO.	CRQA100H07
		REV.:	CN
		PAGE :	4 OF 163

- In the event of Machine Shut Down Notice was issued. QA supervisor ensure to send the Machine Shut Down Notice for acknowledgement within 24 hours / 1 working day. If the machine shut down notice issue during night shift, weekend and public holiday, the machine can be release back to production after attend by engineering and the machine shut down notice can be approved on the next working days.
- To release material physically and electronically through the lot tracking system upon receiving documented and signed MRB.
- To issue MRB to production if found critical defects and 3 times rejection (2 times for automotive product in SZ Site) of the same lot at QA Gate.
- Escalate critical defect as defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19)

5.3.3 QA Engineer / Executive

- To verify the root cause established by process engineer
- To ensure recapture procedure is followed strictly as defined by Process Engineer's Instructions and Specification.
- To evaluate the risk of defect encountered, determine the disposition of non conforming material/products and corrective & preventive actions in MRB, together with process engineer.
- To verify and trigger process owner in order to ensure the MRB procedure is complied with.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19)
- Lot hold under HQ which was raised MRB, once customer provide disposition to scrap the lot and MRB approved, need to change the hold code to HDCRAP and need to notify QA supervisor to Tag and quarantine in the QA cabinet while waiting for the scrap from approval. (Applicable for S-Site Assembly only)

5.3.4 QA Manager

- To decide if customer notification is required or not together with engineering manager.
- To have final decision on the disposition of Non-conforming Material/Products

NOTE: The QA Personnel is authorized to shut down any operation where non-conforming is cited with the approval from the QA manager (Refer to QCG000017)

5.4 Process Engineer/Process Owner

- To inform QA personnel (supervisor/engineer) immediately upon detection of non conforming material
- To stop affected operation/ process/ material when defect is triggered in production line.
- To determine the cause of failure by performing detailed process mapping.
- To identify lots affected and provide list and all related instruction to production / QA for recapture.

Noted:Need to include all customer code during reterive risk lots, such as

Customer	Customer code
TI	TIM,TID,TIG,TIT

- To perform risk assessment of the affected lot to determine the extent of the effect with QA Engineer

CARSEM	PROPRIETARY INFORMATION. DO NOT REPRODUCE OR FURTHER DISCLOSE WITHOUT AUTHORIZATION FROM CARSEM	SPEC NO.	CRQA100H07
		REV.:	CN
		PAGE :	5 OF 163

- To raise MRB, convene a meeting to review the actions necessary to dispose the discrepant parts or lots
- To perform machine/ product buy-off with maintenance before restart production.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16, 17, 18 and 19)

5.5 Engineering Manager

- To ensure Risk Assessment is completed by Engineering and recommend appropriate next course of action on product disposition.
- To decide customer notification for product/ material disposition together with QA manager and lead the communication with customer (if the customer is to be notified).
- To approve the disposition made in MRB after reviewing all pertinent support documents related to the disposition.
- To react to quality alert and ensure corrective & preventive action are taken (As necessary).

5.6 Reliability Engineer

- To perform reliability test for product disposition as per guideline given in the Reliability Test Risk Assessment Matrix (Refer to Attachment C-3).

5.7 PC

- To provide assembly and test WIP report to QA and production in order to perform lot On-hold.

6.0 Material/Equipments: N/A

7.0 Procedure:

7.1 The procedure for quarantine/on-hold, recapture and MRB of non-conforming material/products shall refer to Appendix 1 to Appendix 14:

- 7.1.1 Appendix 1: General Flow for Control of Non-conforming Material/Products. (Critical Defect)
- 7.1.2 Appendix 2: Flow for Control of Incoming Raw Material Failure
- 7.1.3 Appendix 3: Flow for Control of Line Raw Material Failure Feedback
- 7.1.4 Appendix 4: Flow for Control of QA Gate Failure
- 7.1.5 Appendix 5: Flow for Control of Reliability Monitoring Failure
- 7.1.6 Appendix 6: Flow for Control of Customer Complaint & Field Failure
- 7.1.7 Appendix 7: Flow for Control of Assembly Low Yield with Critical Defect Failure
- 7.1.8 Appendix 7A: Flow for Control of Assembly Low Yield/Critical Defect Failure (SZ Site only).
- 7.1.9 Appendix 8: Flow for Control of Metal Finishing Failure
- 7.1.10 Appendix 9: Flow for Control of Lead (Pb) Contamination At Plating Process Failure
- 7.1.11 Appendix 10: Flow for Control of Test & Finishing Failure (Assembly Related Defect)
- 7.1.12 Appendix 11: Flow for Control of Test Low Yield & High Continuity Lot Verification & Failure
- 7.1.13 Appendix 12: Flow for Control of Test and Reel Failure
- 7.1.14 Appendix 13: Flow for Control of measuring equipment / instruments out of tolerance
- 7.1.15 Appendix 14: Flow for Control of Environment out of spec limits
- 7.1.16 Appendix 15: Flow for Control of Non-Conformance Lots at Pre-Test Area